Subject: PROBABILITY & STATISTICS

Subject Code: 23AS4401

UNIT V : Small Sample Tests

Syllabus: Student t-distribution (single mean, two means, paired t-test), F-test for equality of variances, chi-square test for goodness of fit and for independence of attributes.

1. Small Samples and Exact Sampling Distributions

Definition:

A sample is small if $n < 30$. The normal (z) test is not reliable because s is a poor estimate of σ; exact sampling distributions – t, χ^2 and F – are used.

- Assumption: the parent population is (approximately) normal.
- **Degrees of freedom (d.f. = ν):** number of independent observations free to vary = n minus the number of constraints (parameters estimated from the sample).

Test	Distribution	Used for
t-test	Student's t	Means (single, two, paired)
F-test	Snedecor's F	Equality of two variances
Chi-square	χ^2	Goodness of fit, independence

Note:

Same six steps as Unit IV: H0, H1, α, test statistic, table value, conclusion.

2. Student's t-distribution

Definition:

If x1, ..., xn is a random sample from $N(\mu, \sigma^2)$, then $t = (\bar{x} - \mu) / (s/\sqrt{n})$ follows Student's t-distribution with $\nu = n - 1$ degrees of freedom. (W. S. Gosset, "Student", 1908.)

Formula:

p.d.f.: $f(t) = C_\nu (1 + t^2/\nu)^{-(\nu+1)/2}$, $-\infty < t < \infty$

- Also $t = z / \sqrt{(\chi^2/\nu)}$, the ratio of a standard normal to the square root of an independent χ^2 divided by its d.f.

Properties

- Bell shaped and symmetric about $t = 0$ (like normal) but with heavier tails.
- Mean $= 0$ $(\nu > 1)$; Variance $= \nu/(\nu - 2)$ $(\nu > 2) > 1$.
- As $\nu \to \infty$, t-distribution tends to $N(0, 1)$; for $\nu \geq 30$ the normal is a good approximation.
- Shape depends only on ν (one parameter). Table gives $t_\alpha(\nu)$, the point with upper tail area α.
- Odd moments are zero.

d.f. ν	5 % two-tailed	1 % two-tailed	5 % one-tailed
5	2.571	4.032	2.015
7	2.365	3.499	1.895
8	2.306	3.355	1.860
9	2.262	3.250	1.833
10	2.228	3.169	1.812
15	2.131	2.947	1.753
20	2.086	2.845	1.725
24	2.064	2.797	1.711
30	2.042	2.750	1.697
∞	1.960	2.576	1.645

Applications of the t-distribution

- Test for single mean; test for difference of two means (independent samples); paired t-test.
- Test of significance of correlation coefficient; confidence intervals for the mean of a small sample.

3. t-test for a Single Mean

Formula:

H0: $\mu = \mu_0$. $t = (\bar{x} - \mu_0) / (s/\sqrt{n})$, $s^2 = \Sigma(x - \bar{x})^2/(n - 1)$, d.f. $= n - 1$

- If the S.D. is computed with divisor n (call it S), use $t = (\bar{x} - \mu_0)/(S/\sqrt{(n - 1)})$.
- Reject H0 if $|t_{cal}| > t_{tab}(n - 1, \alpha)$.

Confidence limits for μ

Formula:

$\bar{x} \pm t_{\alpha/2}(n - 1) \cdot s/\sqrt{n}$

Example:

Ten values: 50, 52, 48, 51, 49, 53, 47, 50, 52, 48. Test H0: $\mu = 48$ against $\mu \neq 48$ at 5 %.

$\bar{x} = 500/10 = 50$. Deviations: 0, 2, −2, 1, −1, 3, −3, 0, 2, −2; $\Sigma(x - \bar{x})^2 = 36$.

$s^2 = 36/9 = 4 \Rightarrow s = 2$. $t = (50 - 48)/(2/\sqrt{10}) = 2/0.6325 =$ ***3.16***

$t_{tab}(9) = 2.262$. $3.16 > 2.262 \Rightarrow$ ***reject H0***: mean differs from 48.

Example:

For the same data find 95 % confidence limits for μ.

$50 \pm 2.262 \times 0.6325 = 50 \pm 1.43 =$ **(48.57, 51.43)** (48 is outside, consistent with rejection).

Example:

A sample of 25 has mean 102 and s = 10. Test H0: μ = 100 at 5 %.

$t = (102 - 100)/(10/5) = 1$. $t_{tab}(24) = 2.064$. $1 < 2.064 \Rightarrow$ ***accept H0***.

Example:

A machine is to fill 500 g packets. 16 packets give mean 497 g, s = 6 g. Is the machine under-filling? (5 %, left-tailed)

$t = (497 - 500)/(6/4) = -2$. $t_{tab}(15)$ one-tailed 5 % $= -1.753$. $-2 < -1.753 \Rightarrow$ ***reject H0***: under-filling.

Example:

Nine students' marks: mean 56, s = 6 (divisor n – 1). Test μ = 60 at 5 %.

$t = (56 - 60)/(6/3) = -2$. $t_{tab}(8) = 2.306$. $|t| < 2.306 \Rightarrow$ ***accept H0***.

4. t-test for Difference of Two Means (independent samples)

- Assumptions: both populations normal; independent samples; equal variances $\sigma1^2 = \sigma2^2$ (check with F-test first).

Formula:

H0: $\mu_1 = \mu_2$. $t = (\bar{x}_1 - \bar{x}_2) / [\, s\sqrt{(1/n_1 + 1/n_2)}\,]$, d.f. $= n_1 + n_2 - 2$

Pooled variance: $s^2 = [\Sigma(x_1 - \bar{x}_1)^2 + \Sigma(x_2 - \bar{x}_2)^2] / (n_1 + n_2 - 2) = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / (n_1 + n_2 - 2)$

Example:

Sample A: 12, 15, 18, 14, 16. Sample B: 10, 13, 11, 12, 14. Is there a significant difference between means at 5 %?

$\bar{x}_1 = 75/5 = 15$, $\Sigma(x - \bar{x}_1)^2 = 9 + 0 + 9 + 1 + 1 = 20$.

$\bar{x}_2 = 60/5 = 12$, $\Sigma(x - \bar{x}_2)^2 = 4 + 1 + 1 + 0 + 4 = 10$.

$s^2 = (20 + 10)/8 = 3.75 \Rightarrow s = 1.936$. S.E. $= 1.936 \times \sqrt{(2/5)} = 1.2247$

$t = 3/1.2247 =$ ***2.45***. $t_{tab}(8) = 2.306 \Rightarrow 2.45 > 2.306 \Rightarrow$ ***reject H0*** at 5 %.

At 1 % $t_{tab}(8) = 3.355 > 2.45 \Rightarrow$ accept H0 at 1 %.

Example:

Two batches of 10 and 12 items have means 20 and 18 with $s_1 = 3$, $s_2 = 4$. Test equality at 5 %.

$s^2 = (9 \times 9 + 11 \times 16)/20 = (81 + 176)/20 = 12.85$, $s = 3.585$.

S.E. $= 3.585\sqrt{(1/10 + 1/12)} = 3.585 \times 0.4282 = 1.535$. $t = 2/1.535 = 1.30 < t_{tab}(20) = 2.086 \Rightarrow$ ***accept H0***.

Note:

If variances are unequal (F-test rejects) use Welch's approximate t-test – beyond this syllabus.

5. Paired t-test

Definition:

Used when observations are paired (same subjects before and after, matched pairs). The two samples are dependent and have equal size n.

- Let $d_i = x_i - y_i$ (differences). Test whether the mean difference is zero.

Formula:

H0: $\mu d = 0$. $t = \bar{d} / (s_d/\sqrt{n})$, $s_d^2 = \Sigma(d - \bar{d})^2/(n - 1)$, d.f. = n – 1

- It reduces variation between subjects and is more powerful than the independent test when pairs are correlated.

Example:

Marks of 8 students before and after coaching. Test if coaching is effective at 5 % (one-tailed).

Student	1	2	3	4	5	6	7	8
Before	70	72	68	75	80	66	74	71
After	74	75	73	77	86	70	77	76
After – Before	4	3	5	2	6	4	3	5

- $\Sigma d = 32 \Rightarrow \bar{d} = 4$; $\Sigma(d - \bar{d})^2 = 0 + 1 + 1 + 4 + 4 + 0 + 1 + 1 = 12$.
- $s_d^2 = 12/7 = 1.714 \Rightarrow s_d = 1.309$. $t = 4/(1.309/\sqrt{8}) = 4/0.4629 =$ ***8.64***
- $t_{tab}(7)$, one-tailed 5 % = 1.895. $8.64 > 1.895 \Rightarrow$ ***reject H0***: coaching is effective.

Example:

Blood pressure of 5 patients before and after a drug: differences d = 6, 4, 8, 2, 5. Test H0: $\mu d = 0$ (5 %, two-tailed).

$\bar{d} = 5$. $\Sigma(d - \bar{d})^2 = 1 + 1 + 9 + 9 + 0 = 20 \Rightarrow s_d^2 = 5$, $s_d = 2.236$.

$t = 5/(2.236/\sqrt{5}) = 5/1 = 5$. $t_{tab}(4) = 2.776 \Rightarrow 5 > 2.776 \Rightarrow$ ***reject H0***.

6. F-distribution and Test for Equality of Variances

Definition:

If $s1^2$ and $s2^2$ are variances of independent samples from normal populations, then $F = s1^2/s2^2$ (when $\sigma1^2 = \sigma2^2$) follows Snedecor's F-distribution with $(\nu1, \nu2) = (n1 - 1, n2 - 1)$ d.f.

Formula:

$F = (\chi1^2/\nu1) / (\chi2^2/\nu2)$ (ratio of two independent chi-squares each divided by its d.f.)

Properties of the F-distribution

- Positively skewed, defined for $F > 0$; shape depends on $(\nu1, \nu2)$.
- Mean $= \nu2/(\nu2 - 2)$ for $\nu2 > 2$.
- Reciprocal property: $F_{1-\alpha}(\nu1, \nu2) = 1 / F_{\alpha}(\nu2, \nu1)$.
- If t has ν d.f. then t^2 has $F(1, \nu)$ distribution.

F-test procedure

Formula:

$H0: \sigma1^2 = \sigma2^2$. F = (larger variance) / (smaller variance) $= s_L^2 / s_S^2 \geq 1$

- d.f. $= (n_L - 1, n_S - 1)$, where n_L is the size of the sample with the larger variance.
- Reject H0 if $F_{cal} > F_{tab}(\nu1, \nu2)$ at the chosen α (for a two-tailed test use $\alpha/2$ table value).

Some F values (5 %, upper)	$(\nu1, \nu2)$
3.29	(7, 9)
3.35	(10, 8)
2.98	(10, 10)

5.05	(5, 5)
6.39	(4, 4)

Example:

Sample 1: $n = 8$, $\Sigma(x - \bar{x})^2 = 84$. Sample 2: $n = 10$, $\Sigma(x - \bar{x})^2 = 45$. Test equality of variances at 5 % (one-tailed).

$s1^2 = 84/7 = 12$; $s2^2 = 45/9 = 5$. $F = 12/5 = \mathbf{2.4}$, d.f. (7, 9).

$F_{tab}(7, 9) = 3.29$. $2.4 < 3.29 \Rightarrow$ ***accept H0***: variances may be taken equal.

Example:

$n1 = 11$, $s1^2 = 25$; $n2 = 9$, $s2^2 = 16$. Test $\sigma1^2 = \sigma2^2$ at 5 %.

$F = 25/16 = 1.5625$, d.f. (10, 8), $F_{tab} = 3.35 \Rightarrow$ ***accept H0***.

Note:

Do the F-test first; if variances are equal, then use the pooled t-test for two means.

7. Chi-square Distribution

Definition:

If $Z1, \ldots, Z\nu$ are independent standard normal variables then $\chi^2 = Z1^2 + \ldots + Z\nu^2$ follows chi-square distribution with ν d.f.

Formula:

p.d.f.: $f(\chi^2) = C \cdot e^{-\chi^2/2} (\chi^2)^{\nu/2 - 1}$, $\chi^2 > 0$

Properties

- Mean = ν; Variance = 2ν; Mode = $\nu - 2$ ($\nu \geq 2$).

- Positively skewed; tends to normal $N(\nu, 2\nu)$ for large ν. χ^2 is never negative.
- Additive: χ^2 with $\nu1$ d.f. plus independent χ^2 with $\nu2$ d.f. gives χ^2 with $\nu1 + \nu2$ d.f.
- For a normal sample: $(n - 1)s^2/\sigma^2 \sim \chi^2(n - 1)$ (basis for variance tests).

d.f.	5 % point	1 % point
1	3.841	6.635
2	5.991	9.210
3	7.815	11.345
4	9.488	13.277
5	11.070	15.086
6	12.592	16.812
8	15.507	20.090
10	18.307	23.209

Conditions for applying χ^2 tests

- Total frequency N is large (at least 50) and observations are independent.
- Each expected frequency ≥ 5; if smaller, pool neighbouring classes (and reduce d.f. accordingly).
- Frequencies are actual counts, not percentages.

8. χ^2 Test for Goodness of Fit

Definition:

Tests whether observed frequencies agree with those expected under a theoretical distribution (uniform, binomial, Poisson, normal, 9:3:3:1 ratio, …).

Formula:

H0: the data fit the distribution. $\chi^2 = \Sigma (O - E)^2 / E$

d.f. = k – 1 – m (k = number of classes, m = number of parameters estimated from the data)

- Reject H0 if $\chi^2_{cal} > \chi^2_{tab}(d.f., \alpha)$. It is always a right-tailed test.

Example:

A die is thrown 120 times with outcomes 1 to 6 observed as 15, 25, 20, 18, 22, 20. Is it fair (5 %)?

Face	1	2	3	4	5	6
O	15	25	20	18	22	20
E	20	20	20	20	20	20
$(O - E)^2/E$	1.25	1.25	0	0.2	0.2	0

- $\chi^2 = 1.25 + 1.25 + 0 + 0.2 + 0.2 + 0 = \mathbf{2.9}$; d.f. = 5; $\chi^2_{tab} = 11.07 \Rightarrow 2.9 < 11.07 \Rightarrow$ ***accept H0***: die is fair.

Example:

Mendel: peas in the ratio 9:3:3:1 expected. Observed 315, 108, 101, 32 (N = 556). Test at 5 %.

Class	O	E (9:3:3:1)	$(O - E)^2/E$
Round yellow	315	312.75	0.0162
Round green	108	104.25	0.1349
Wrinkled yellow	101	104.25	0.1013
Wrinkled green	32	34.75	0.2176

- $\chi^2 = 0.0162 + 0.1349 + 0.1013 + 0.2176 = \mathbf{0.470}$; d.f. = 3; $\chi^2_{tab} = 7.815 \Rightarrow$ ***accept H0***; data fit the 9:3:3:1 ratio very well.

Example:

Fitting a binomial: d.f. = k – 1 – 1 = k – 2 if p is estimated from data; k – 1 if p is given.

Example:

200 families with 4 children each; observed boys 0 to 4: 12, 52, 76, 44, 16. Do they fit B(4, 1/2)? (5 %)

Expected = 200 × (1, 4, 6, 4, 1)/16 = 12.5, 50, 75, 50, 12.5.

$\chi^2 = 0.5^2/12.5 + 2^2/50 + 1/75 + 6^2/50 + 3.5^2/12.5 = 0.02 + 0.08 + 0.0133 + 0.72 + 0.98 =$ ***1.813***

d.f. = 4 (p given), $\chi^2_{tab} = 9.488 \Rightarrow$ ***accept H0***: binomial fit is acceptable.

9. χ^2 Test for Independence of Attributes

Definition:

Tests whether two attributes (classifications) A and B are independent, using an r × c contingency table.

Formula:

H0: attributes A and B are independent.

Expected frequency E(ij) = (row total i × column total j) / N

$\chi^2 = \Sigma (O - E)^2 / E$, d.f. = (r – 1)(c – 1)

Procedure

- Step 1: Write the observed table with row and column totals. Step 2: compute each expected frequency.
- Step 3: compute χ^2. Step 4: compare with χ^2_{tab} with (r – 1)(c – 1) d.f. Step 5: conclude.

Shortcut for a 2 × 2 table [a b / c d], N = a + b + c + d

Formula:

$\chi^2 = N(ad - bc)^2 / [(a + b)(c + d)(a + c)(b + d)]$, d.f. = 1

Yates' correction (if any cell < 5): $\chi^2 = N(|ad - bc| - N/2)^2 / [(a + b)(c + d)(a + c)(b + d)]$

Example:

Recovery and treatment. Test if recovery is independent of treatment (5 %).

Treatment	Recovered	Not recovered	Total
A	40	10	50
B	30	20	50
Total	70	30	100

- Expected: A: 35, 15; B: 35, 15. $\chi^2 = 25/35 + 25/15 + 25/35 + 25/15 =$ **4.76**
- Check by shortcut: $100(800 - 300)^2/(50 \times 50 \times 70 \times 30) = 25{,}000{,}000/5{,}250{,}000 = 4.76$ ✓
- d.f. = 1, $\chi^2_{tab} = 3.841 \Rightarrow 4.76 > 3.841 \Rightarrow$ ***reject H0***: recovery depends on treatment.
- With Yates: $100(500 - 50)^2/5{,}250{,}000 = 3.857 > 3.841 \Rightarrow$ still reject (barely).

Example:

Gender and preference for three products P, Q, R. Test independence at 5 %.

Gender	P	Q	R	Total
Men	30	40	30	100
Women	20	30	50	100
Total	50	70	80	200

- Expected: Men 25, 35, 40; Women 25, 35, 40.
- $\chi^2 = 25/25 + 25/35 + 100/40 + 25/25 + 25/35 + 100/40 = 1 + 0.714 + 2.5 + 1 + 0.714 + 2.5 =$ ***8.43***
- d.f. = (2 – 1)(3 – 1) = 2, $\chi^2_{tab} = 5.991 \Rightarrow 8.43 > 5.991 \Rightarrow$ ***reject H0***: preference depends on gender.

> ***Note:***
> Independence test: do ***not*** use percentages; and the test says "associated", not "caused by".

10. Practice Problems with Answers

- 1) A sample of 16 has $\bar{X} = 53$ and $s = 4$. Test $\mu = 50$ at 5 %. Ans: $t = 3/(4/4) = 3 > 2.131 \Rightarrow$ reject.
- 2) Two samples $n1 = 7$, $\bar{X}1 = 28$, $s1^2 = 6$; $n2 = 9$, $\bar{X}2 = 24$, $s2^2 = 8$. Test equal means. Ans: pooled $s^2 = (36 + 64)/14 = 7.14$, S.E. $= 2.672 \times 0.504 = 1.347$, $t = 2.97 > 2.145 \Rightarrow$ reject.
- 3) Test H0: σ^2 equal for $s1^2 = 20$ ($n = 13$), $s2^2 = 10$ ($n = 9$). Ans: $F = 2$, d.f. (12, 8), $F_{tab} \approx 3.28 \Rightarrow$ accept.
- 4) 4 categories observed 30, 20, 25, 25 ($N = 100$); equal probabilities? Ans: $E = 25$, $\chi^2 = (25 + 25 + 0 + 0)/25 = 2 < 7.815 \Rightarrow$ accept.
- 5) A 2 × 2 table 20 30 / 40 10: find χ^2. Ans: $N = 100$, $100(200 - 1200)^2/(50 \times 50 \times 60 \times 40) = 100 \times 10^6/6 \times 10^6 = 16.67 \Rightarrow$ reject.
- 6) Why must the F ratio be written with larger variance on top? Ans: so $F \geq 1$ and only upper table values are needed.
- 7) State the properties of the t-distribution and chi-square distribution.
- 8) Distinguish independent t-test and paired t-test.

11. More Worked Examples – t-tests

Example:

Six observations: 2, 4, 6, 8, 10, 12. Test H0: $\mu = 5$ at 5 %.

$\bar{x} = 7$; $\Sigma(x - \bar{x})^2 = 25 + 9 + 1 + 1 + 9 + 25 = 70 \Rightarrow s^2 = 14$, $s = 3.742$.

$t = (7 - 5)/(3.742/\sqrt{6}) = 2/1.5275 = \mathbf{1.31}$; $t_{tab}(5) = 2.571 \Rightarrow$ ***accept H0***.

Example:

Sample C: 22, 24, 26, 28. Sample D: 18, 20, 22, 24, 26. Test equality of means (5 %).

$\bar{x}1 = 25$, $SS1 = 9 + 1 + 1 + 9 = 20$; $\bar{x}2 = 22$, $SS2 = 16 + 4 + 0 + 4 + 16 = 40$.

$s^2 = 60/7 = 8.571$, $s = 2.928$; S.E. $= 2.928\sqrt{(1/4 + 1/5)} = 1.964$.

$t = 3/1.964 = \mathbf{1.53}$; $t_{tab}(7) = 2.365 \Rightarrow$ ***accept H0***.

Example:

Weights (kg) of 6 persons reduce after a diet by d = 3, 5, 2, 4, 6, 4. Is the diet effective? (5 %, one-tailed)

$\bar{d} = 4$; $\Sigma(d - \bar{d})^2 = 1 + 1 + 4 + 0 + 4 + 0 = 10 \Rightarrow s_d^2 = 2$, $s_d = 1.414$.

$t = 4/(1.414/\sqrt{6}) = 4/0.577 = \mathbf{6.93}$; $t_{tab}(5)$ one-tailed $= 2.015 \Rightarrow$ ***reject H0***: diet effective.

Test of significance of a correlation coefficient

Formula:

H0: $\rho = 0$. $t = r\sqrt{(n - 2)} / \sqrt{(1 - r^2)}$, d.f. $= n - 2$

Example:

$r = 0.6$ from $n = 27$ pairs. Is it significant at 5 %?

$t = 0.6 \times 5/\sqrt{(1 - 0.36)} = 3/0.8 = \mathbf{3.75}$; $t_{tab}(25) = 2.060 \Rightarrow$ ***significant***.

Test for a single variance (uses χ^2)

Formula:

H0: $\sigma^2 = \sigma_0^2$. $\chi^2 = (n-1)s^2 / \sigma_0^2$, d.f. = $n-1$

Example:

$n = 10$, $s^2 = 6.4$. Test H0: $\sigma^2 = 4$ at 5 % (right-tailed).

$\chi^2 = 9 \times 6.4/4 = \mathbf{14.4}$; $\chi^2_{tab}(9) = 16.919 \Rightarrow 14.4 < 16.919 \Rightarrow$ ***accept H0***.

12. More Worked Examples – χ^2 Tests

Example:

Genetic ratio 1:2:1 expected. Observed 18, 45, 17 (N = 80). Test at 5 %.

Expected = 20, 40, 20. $\chi^2 = 4/20 + 25/40 + 9/20 = 0.2 + 0.625 + 0.45 =$ ***1.275***

d.f. = 2, $\chi^2_{tab} = 5.991 \Rightarrow$ ***accept H0***.

Example:

Accidents on Mon–Fri were 14, 18, 12, 11, 15 (total 70). Are accidents uniformly spread? (5 %)

Expected = 14 each. $\chi^2 = (0 + 16 + 4 + 9 + 1)/14 = 30/14 =$ ***2.14***

d.f. = 4, $\chi^2_{tab} = 9.488 \Rightarrow$ ***accept H0***: no day effect.

Pooling classes

- If an expected frequency is below 5, combine it with the neighbouring class and recalculate d.f. using the new number of classes.

Example:

Vaccination and attack: test independence at 5 %.

Group	Not attacked	Attacked	Total
Vaccinated	85	15	100
Not vaccinated	60	40	100
Total	145	55	200

- $ad - bc = 85 \times 40 - 15 \times 60 = 3400 - 900 = 2500$.
- $\chi^2 = 200 \times 2500^2 / (100 \times 100 \times 145 \times 55) = 1{,}250{,}000{,}000/79{,}750{,}000 = \mathbf{15.67}$
- d.f. = 1, $\chi^2_{tab} = 3.841 \Rightarrow$ ***reject H0***: vaccination is associated with protection.

Example:

A 3 × 3 table (income level vs. region). Test independence at 5 %.

Region	Low	Medium	High	Total
A	20	30	10	60
B	30	40	30	100
C	10	20	10	40
Total	60	90	50	200

- Expected: A: 18, 27, 15; B: 30, 45, 25; C: 12, 18, 10.
- $\chi^2 = 0.222 + 0.333 + 1.667 + 0 + 0.556 + 1 + 0.333 + 0.222 + 0 = \mathbf{4.33}$
- d.f. = $(3 - 1)(3 - 1) = 4$, $\chi^2_{tab} = 9.488 \Rightarrow 4.33 < 9.488 \Rightarrow$ ***accept H0***: independent.

13. Summary of All Tests

Test	Statistic	d.f.
Large sample mean	$z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$	–
Large sample proportion	$z = (p - P_0)/\sqrt{(P_0Q_0/n)}$	–

Single mean (small)	$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$	$n - 1$
Two means (small)	$t = (\bar{x}_1 - \bar{x}_2)/[s\sqrt{(1/n_1 + 1/n_2)}]$	$n_1 + n_2 - 2$
Paired	$t = \bar{d}/(s_d/\sqrt{n})$	$n - 1$
Two variances	$F = s_1^2/s_2^2$ (larger/smaller)	$(n_1 - 1, n_2 - 1)$
Single variance	$\chi^2 = (n - 1)s^2/\sigma_0^2$	$n - 1$
Goodness of fit	$\chi^2 = \Sigma(O - E)^2/E$	$k - 1 - m$
Independence	$\chi^2 = \Sigma(O - E)^2/E$	$(r - 1)(c - 1)$

Choosing a test (decision guide)

- Testing means or proportions with $n \geq 30$ → z test (Unit IV).
- Testing means with $n < 30$ and σ unknown → t-test: one sample / two independent samples / paired.
- Comparing the spread of two samples → F-test.
- Counts in categories vs a theory, or two categorical attributes → χ^2 test.

> **Note:**
> Under H0 the statistics t, F and χ^2 are all linked to the normal distribution: $t = z/\sqrt{(\chi^2/\nu)}$, F = ratio of χ^2's.

14. More Practice Problems

- 9) A sample of 10 has $s^2 = 5$ and a second sample of 13 has $s^2 = 3$. Test equal variances. Ans: $F = 5/3 = 1.67$, d.f. (9, 12), $F_{tab} \approx 2.80 \Rightarrow$ accept.
- 10) Heights of 12 men have mean 170 cm, $s = 4$. Test $\mu = 168$ at 5 %. Ans: $t = 2/(4/\sqrt{12}) = 1.73 < 2.201 \Rightarrow$ accept.
- 11) Observed counts 40, 60 and expected 50, 50. χ^2? Ans: $(100 + 100)/50 = 4 > 3.841 \Rightarrow$ reject at 5 % (d.f. 1).
- 12) Why do we use $n - 1$ as d.f. for a single-sample t-test? Ans: one parameter ($\bar{x}$) is estimated from the sample.

- 13) Give 99 % confidence limits for μ from $n = 9$, $\bar{x} = 20$, $s = 3$. Ans: $20 \pm 3.355 \times 1 = (16.65, 23.36)$.
- 14) For χ^2 test of independence in a 4×3 table what are the d.f.? Ans: $3 \times 2 = 6$.

Quick Revision – Unit V

- Single mean: $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$, d.f. $n - 1$.
- Two means: $t = (\bar{x}_1 - \bar{x}_2)/[s\sqrt{(1/n_1 + 1/n_2)}]$, d.f. $n_1 + n_2 - 2$.
- Paired: $t = \bar{d}/(s_d/\sqrt{n})$, d.f. $n - 1$.
- F = larger s^2 / smaller s^2, d.f. $(n_L - 1, n_S - 1)$.
- $\chi^2 = \Sigma(O - E)^2/E$; fit d.f. $= k - 1 - m$; independence d.f. $= (r - 1)(c - 1)$.
- $E(ij) = (\text{row total} \times \text{column total})/N$.